Activity #4 Use High-Low and Regression to Analyze Costs

Bella Corporation is trying to predict its manufacturing overhead costs for the upcoming year; they are debating the use of the high-low method versus the use of regression analysis. They have gathered information about their manufacturing overhead costs in each of the past six months. A table containing their cost data and the associated machine hours in each month (the cost driver) follows.

Month	Manufacturing overhead costs	Number of machine hours
April	$17,000	14,250
May	$14,500	12,010
June	$13,500	10,280
July	$15,000	11,910
August	$13,200	11,080
September	$14,500	11,670

The company performed a regression analysis using the above data and had the following results. (Note: the results are excerpts so not all of the regression analysis results are presented.)

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.980598097
R Square	0.961572627
Adjusted R Square	0.951965784
Standard Error	293.5325786
Observations	6

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	8624104.501	8624104.501	100.0924664	0.000560999
Residual	4	344645.4989	86161.37471		
Total	5	8968750			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*
Intercept	2929.07161	1175.178248	2.492448796	0.067307822	-333.7462853
X Variable 1	0.985611943	0.098515658	10.00462225	0.000560999	0.712088627

a. What is the cost equation if the high-low method is used to estimate costs?

b. Using the high-low method, predict total manufacturing overhead costs if Bella Corporation uses 11,000 hours.

c. What is the cost equation if regression analysis is used to estimate costs (use the results from the regression analysis provided)?

d. Using the results from the regression analysis provided, predict total manufacturing overhead costs if Bella Corporation uses 11,000 hours.

e. Which method (high-low or regression analysis) is a better predictor of total manufacturing overhead costs? Why?